

Looping:

With JavaScript you can use two types of loops, the 'for' loop and the 'while' loop. They can both do the same thing, but in different ways. The 'for' loop is more suited when one needs to iterate over fixed values, such as looping over all the paragraphs in a document and setting them to bold.

Let us see an example where we use the for loop to add the first 100 numbers:

```
sum = 0;
for (i = 1; i <= 100; i++) {
    sum += i;
}
```

The 'for' loop has three parts in the bracket:

- In the first part, you set any variables you need to, this is performed only once at the beginning of the loop.
- In the second part, you test the condition for the execution of the loop, this is tested before the loop is executed, so if the statement is false to begin with, the loop will never execute.
- The third part performs operations that change the counter, incrementing or decrementing it.
- The statements in the loop are executed every time the loop is run.

Now let's see the 'while' loop doing the same:

```
while (cond==true)
{
    ...statements...
}
```

Doing the same using the for loop would be unnecessarily confusing. The 'while' loop shows its strengths here.

Functions:

Functions are a feature present in almost each and every programming language. For anyone who has studied mathematics, understanding functions is not conceptually difficult. However, practically, computer functions are different and more versatile. Let us see an example of a function which returns the square of an entered number: